

MAKEEV'S REGULAR-SIMPLEX CONJECTURE: POSITIVE RESULTS AND A COUNTEREXAMPLE IN DIMENSION FOUR

WORKING DRAFT

ABSTRACT. Let

$$V = \mathbf{1}^\perp \subset \mathbb{R}^{d+1}, \quad u_{ij} = \frac{e_i - e_j}{\sqrt{2}} \in S(V), \quad 0 \leq i < j \leq d,$$

be the normalized roots of type A_d . We study whether every smooth odd $f: S(V) \rightarrow \mathbb{R}$ agrees with a linear functional on a rotated copy of these roots. A single variational identity identifies the cycle defect of a cubic with the gradient of its correlation with $\sum x_i^3$. It yields the desired conclusion for all spherical harmonics of degrees 1 and 3, for a nonempty C^0 -open set of arbitrary odd functions, and for all zonal odd functions.

In dimension 4 we then give an explicit counterexample. Identifying $\mathbb{R}^4 \cong \mathbb{C}^2$, for every sufficiently small nonzero real ε the function

$$\operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon \left(|z_1|^2 - \frac{3}{5} \right) \operatorname{Im}(z_1^2 \bar{z}_2), \quad (z_1, z_2) \in S^3,$$

does not agree with any linear functional on any rotated A_4 root system. The two summands are spherical harmonics of degrees 3 and 5. The main proof is reduced to a symmetry identity, a compact obstruction lemma, and one explicit algebraic transversality proposition; the complete algebraic verification is placed in an appendix.

1. INTRODUCTION

Makeev conjectured that the dual difference body of a regular simplex is a universal cover for sets of diameter one [7]. Kuperberg showed that the constant-width formulation is equivalent to the following function-theoretic assertion: every continuous odd function on S^{d-1} agrees with a linear function on a rotated copy of the A_d root system. He proved the assertion for $d = 3$ and showed that the homological obstruction used there vanishes in every higher dimension [6]. A related equivariant proof in dimension three was given by Hausel, Makai, and Szűcs [3]. Chen's July 2026 preprint still records the regular- A_d universal-cover statement as Makeev's conjecture [1].

We normalize the roots throughout:

$$V = \mathbf{1}^\perp \subset \mathbb{R}^m, \quad m = d + 1, \quad u_{ij} = \frac{e_i - e_j}{\sqrt{2}} \in S(V). \quad (1)$$

This normalization is essential because the function is defined on the unit sphere. The root configuration has an orientation-reversing symmetry, for example the restriction to V of a transposition of two coordinate vectors. Consequently, replacing $O(V)$ by $SO(V)$ does not change the problem.

The first part of the paper records three positive results. The main device is a gradient identity for cubics, closely related in spirit to variational orthogonal tensor diagonalization [8]. The second part gives a negative answer in dimension four. Its perturbative structure is similar to the compactness-plus-infinitesimal-obstruction strategy that appears in Karasev's counterexample

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This manuscript reorganizes and simplifies the supplied arguments. It makes no claim of priority.

to another conjecture of Makeev [5], although the obstruction here is different and completely explicit.

Theorem 1.1 (Main counterexample). *There is $\varepsilon_0 > 0$ such that, for every real ε with $0 < |\varepsilon| < \varepsilon_0$, the smooth odd function*

$$\tilde{f}_\varepsilon(z_1, z_2) = \operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon \left(|z_1|^2 - \frac{3}{5} \right) \operatorname{Im}(z_1^2 \bar{z}_2) \quad (2)$$

on $S^3 \subset \mathbb{C}^2$ has no $A \in \operatorname{SO}(4)$ and $b \in \mathbb{R}^4$ satisfying

$$\tilde{f}_\varepsilon(A\nu_{ij}) = b \cdot A\nu_{ij} \quad (0 \leq i < j \leq 4).$$

2. EDGE FLOWS AND THE TEST MAP

Let $\mathcal{E}$ be the space of skew edge labels on the complete graph K_m :

$$\mathcal{E} = \{Y \in \mathbb{R}^{m \times m} : Y^T = -Y\}, \quad \langle X, Y \rangle_{\mathcal{E}} = \sum_{i < j} X_{ij} Y_{ij}.$$

For $c \in V$, define

$$(\delta c)_{ij} = \frac{c_i - c_j}{\sqrt{2}}.$$

These are exactly the restrictions of linear functions on V to the roots (1). Put

$$\mathcal{C} = \{X \in \mathcal{E} : X\mathbf{1} = 0\}. \quad (3)$$

This is the divergence-free or cycle space. The orthogonal splitting $\mathcal{E} = \delta V \oplus \mathcal{C}$ is the complete-graph instance of combinatorial Hodge theory for edge flows [4].

Lemma 2.1 (Exact edge labels). *For $Y \in \mathcal{E}$, the following are equivalent.*

(1) *There is $c \in V$ such that*

$$Y_{ij} = \frac{c_i - c_j}{\sqrt{2}}.$$

(2) *Every triangle sum vanishes:*

$$Y_{ij} + Y_{jk} + Y_{ki} = 0.$$

(3) *Y is orthogonal to $\mathcal{C}$.*

(4) *With $\Pi = I - m^{-1}\mathbf{1}\mathbf{1}^T$, one has*

$$\Pi Y \Pi = 0.$$

Proof. The first condition implies the second by telescoping. Conversely, if all triangle sums vanish, set $c_0 = 0$ and $c_i = \sqrt{2}Y_{i0}$. The triangle $(i, j, 0)$ gives $Y_{ij} = (c_i - c_j)/\sqrt{2}$; subtracting the mean of the c_i places c in V .

If $X\mathbf{1} = 0$, then

$$\sum_{i < j} X_{ij}(c_i - c_j) = \sum_i c_i \sum_j X_{ij} = 0,$$

so exact labels are orthogonal to $\mathcal{C}$. The divergence map $X \mapsto X\mathbf{1}$ maps $\mathcal{E}$ onto V : for $v \in V$, the skew matrix $X_{ij} = v_i - v_j$ satisfies $X\mathbf{1} = mv$. Hence

$$\dim \mathcal{C} = \binom{m}{2} - (m - 1),$$

and the orthogonal complement of $\mathcal{C}$ has the same dimension as δV . This proves the equivalence of the first three conditions.

For the matrix criterion, an exact label has the form $Y = \rho \mathbf{1}^T - \mathbf{1} \rho^T$, so $\Pi Y \Pi = 0$. Conversely, suppose $\Pi Y \Pi = 0$, and write $J_0 = m^{-1} \mathbf{1} \mathbf{1}^T$. Since Y is skew, $J_0 Y J_0 = 0$, and hence

$$Y = J_0 Y + Y J_0.$$

If $\rho_i = m^{-1} \sum_j Y_{ij}$, then $(Y J_0)_{ij} = \rho_i$ and $(J_0 Y)_{ij} = -\rho_j$. Thus $Y_{ij} = \rho_i - \rho_j$, which is exact after taking $c_i = \sqrt{2} \rho_i$ and subtracting the mean. $\square$

The space $\mathcal{C}$ is naturally $\mathfrak{so}(V)$: extend a skew operator on V by zero on $\mathbb{R} \mathbf{1}$. For an odd continuous function $f: S(V) \rightarrow \mathbb{R}$ and $A \in \text{SO}(V)$, define

$$Y_f(A)_{ij} = f(A\nu_{ij}), \quad \mathcal{T}_f(A) = \text{proj}_{\mathcal{C}} Y_f(A). \quad (4)$$

Corollary 2.2 (Test-map formulation). *There are $A \in \text{SO}(V)$ and $b \in V$ such that*

$$f(A\nu_{ij}) = b \cdot A\nu_{ij} \quad (i < j)$$

if and only if $\mathcal{T}_f(A) = 0$ for some $A \in \text{SO}(V)$.

Proof. If $\mathcal{T}_f(A) = 0$, then [theorem 2.1](#) gives $c \in V$ with $f(A\nu_{ij}) = c \cdot \nu_{ij}$. Taking $b = Ac$ gives the stated identity. The converse is immediate. $\square$

Notice the dimension equality

$$\dim \text{SO}(d) = \binom{d}{2} = \dim \mathcal{C}.$$

It is the source of the degree and Euler-class approaches in the earlier literature [6, 3].

3. THE CUBIC GRADIENT IDENTITY

Define the distinguished cubic

$$p(x) = \sum_{i=0}^d x_i^3, \quad x = (x_0, \dots, x_d) \in V. \quad (5)$$

If Q is a cubic on V , let T_Q be its symmetric trilinear polarization, so that $Q(x) = T_Q(x, x, x)$. Extend T_Q to $\mathbb{R}^m$ by projecting each argument to V , and use the invariant tensor inner product

$$\langle Q, R \rangle = \sum_{a,b,c=0}^d (T_Q)_{abc} (T_R)_{abc}.$$

The group acts on cubics by $(g \cdot Q)(x) = Q(g^{-1}x)$.

Lemma 3.1 (Cubic edge identity). *For every cubic Q on V and every $X \in \mathcal{C} = \mathfrak{so}(V)$,*

$$\langle Q, X \cdot p \rangle = 2\sqrt{2} \sum_{i < j} X_{ij} Q(\nu_{ij}). \quad (6)$$

Proof. The infinitesimal action gives

$$(X \cdot p)(x) = \left. \frac{d}{dt} \right|_{t=0} p(e^{-tX} x) = -3 \sum_{i,j} X_{ij} x_i^2 x_j,$$

so

$$\langle Q, X \cdot p \rangle = -3 \sum_{i,j} X_{ij} (T_Q)_{iij}. \quad (7)$$

On the other hand,

$$Q(e_i - e_j) = (T_Q)_{iii} - 3(T_Q)_{iij} + 3(T_Q)_{ijj} - (T_Q)_{jjj}.$$

After multiplying by X_{ij} and summing, the pure diagonal terms vanish because $X\mathbf{1} = 0$. Interchanging i and j in the remaining second term shows

$$\sum_{i < j} X_{ij} Q(e_i - e_j) = -3 \sum_{i, j} X_{ij} (T_Q)_{ij}.$$

Comparison with (7), followed by $Q(e_i - e_j) = 2\sqrt{2}Q(\nu_{ij})$, proves the identity. $\square$

For a cubic Q , set

$$\Psi_Q(A) = \langle Q, A \cdot p \rangle, \quad A \in \mathrm{SO}(V).$$

Identify $T_A \mathrm{SO}(V)$ with $\{AX : X \in \mathfrak{so}(V)\}$ using the curve Ae^{tX} .

Proposition 3.2 (Gradient form of the test map). *For every $A \in \mathrm{SO}(V)$ and $X \in \mathcal{C}$,*

$$d\Psi_Q(A)[AX] = 2\sqrt{2} \langle \mathcal{T}_Q(A), X \rangle_{\mathcal{E}}. \quad (8)$$

Proof. By invariance of the tensor inner product and [theorem 3.1](#),

$$\begin{aligned} d\Psi_Q(A)[AX] &= \langle Q, A \cdot (X \cdot p) \rangle = \langle A^{-1} \cdot Q, X \cdot p \rangle \\ &= 2\sqrt{2} \sum_{i < j} X_{ij} Q(A\nu_{ij}) = 2\sqrt{2} \langle \mathcal{T}_Q(A), X \rangle_{\mathcal{E}}. \end{aligned}$$

The last equality uses $X \in \mathcal{C}$, so projection onto $\mathcal{C}$ does not change the pairing. $\square$

Theorem 3.3 (Cubics plus linear functions). *Let $d \geq 2$. If*

$$f(x) = Q(x) + \ell(x), \quad x \in S(V),$$

where Q is homogeneous cubic and ℓ is linear, then there are $A \in \mathrm{SO}(V)$ and $b \in V$ such that

$$f(A\nu_{ij}) = b \cdot A\nu_{ij} \quad (i < j).$$

Equivalently, the conclusion holds for every odd function whose spherical harmonic expansion has only degrees 1 and 3.

Proof. The compact group $\mathrm{SO}(V)$ contains a critical point A of Ψ_Q . By (8), $\mathcal{T}_Q(A) = 0$. Thus [theorem 2.1](#) gives $c \in V$ such that $Q(A\nu_{ij}) = c \cdot \nu_{ij}$. If $\ell(x) = a \cdot x$, then with $b = Ac + a$,

$$b \cdot A\nu_{ij} = Q(A\nu_{ij}) + \ell(A\nu_{ij}).$$

$\square$

We will also use a coordinate-free form. Let $q_0, \dots, q_d \in \mathbb{R}^d$ be the vertices of a centered regular simplex with

$$\langle q_i, q_j \rangle = \delta_{ij} - \frac{1}{d+1}, \quad u_{ij} = \frac{q_i - q_j}{\sqrt{2}},$$

and put $p_q(x) = \sum_i \langle q_i, x \rangle^3$. Transporting (6) by an isometry gives

$$\langle C, K \cdot p_q \rangle = 2\sqrt{2} \sum_{i < j} \langle q_i, Kq_j \rangle C(\nu_{ij}). \quad (9)$$

Corollary 3.4 (Annihilation from a continuous symmetry). *Let $h_\theta = e^{\theta J} \subset \mathrm{SO}(d)$ be a one-parameter subgroup, and let C be a cubic satisfying $C \circ h_\theta = C$. For $g \in \mathrm{SO}(d)$ define*

$$X_C(g)_{ij} = \langle gq_i, Jgq_j \rangle, \quad \Lambda_g(Y) = \sum_{i < j} X_C(g)_{ij} Y_{ij}.$$

Then

$$\Lambda_g((C(g\nu_{ij}))_{ij}) = 0 \quad \text{for every } g \in \mathrm{SO}(d). \quad (10)$$

Proof. The function $\Psi_C(g) = \langle C, g \cdot p_q \rangle$ is invariant under left multiplication by h_θ . At g , the corresponding tangent vector is $Jg = gK$, where $K = g^{-1}Jg$. Applying (9) to the derivative in this direction yields

$$0 = 2\sqrt{2} \sum_{i < j} \langle q_i, Kq_j \rangle C(g\nu_{ij}) = 2\sqrt{2} \sum_{i < j} \langle gq_i, Jgq_j \rangle C(g\nu_{ij}).$$

□

4. TWO FURTHER POSITIVE RESULTS

4.1. A C^0 -open set of arbitrary odd functions.

Theorem 4.1 (Uniform stability near p). *For every $d \geq 2$ there exists $\eta_d > 0$ such that every continuous odd $f: S(V) \rightarrow \mathbb{R}$ satisfying*

$$\|f - p\|_\infty < \eta_d$$

has a zero of its test map. The same is true near every nonzero scalar multiple or rotation of p , and adding a linear function does not affect the conclusion.

Proof. Since $p(\nu_{ij}) = 0$, one has $\mathcal{T}_p(I) = 0$. For $X \in \mathcal{C} = \mathfrak{so}(V)$ and $u = \nu_{ij}$,

$$dp_u(Xu) = 3 \sum_k u_k^2 (Xu)_k = \frac{3}{2} ((Xu)_i + (Xu)_j).$$

Now

$$(Xu)_i = (Xu)_j = -\frac{X_{ij}}{\sqrt{2}},$$

and hence

$$D\mathcal{T}_p(I)[X] = -\frac{3}{\sqrt{2}}X. \tag{11}$$

Thus I is a nondegenerate zero. By the inverse function theorem, choose a small exponential-coordinate ball $B \subset \mathfrak{so}(V)$ on which this is the only zero. Its local Brouwer degree is the sign of the determinant in (11), and is therefore nonzero. Standard homotopy invariance and stability of local degree under uniform perturbations then apply [2].

For completeness, if $N = \binom{d+1}{2}$, then uniformly in A ,

$$\|\mathcal{T}_f(A) - \mathcal{T}_p(A)\| \leq \sqrt{N} \|f - p\|_\infty.$$

Taking $\|f - p\|_\infty$ smaller than the distance of $\mathcal{T}_p(e^X)$ from 0 on ∂B , divided by $2\sqrt{N}$, keeps the entire straight-line homotopy away from 0 on ∂B . Nonzero degree therefore forces a zero of $\mathcal{T}_f$ in B . □

4.2. Zonal odd functions.

Theorem 4.2 (Zonal functions). *Let $d \geq 2$, $a \in S(V)$, and let $h: [-1, 1] \rightarrow \mathbb{R}$ be continuous and odd. Then every function*

$$f(x) = h(a \cdot x) + \ell(x),$$

with ℓ linear, satisfies the desired conclusion.

Proof. It is enough to take $\ell = 0$. Choose a nontrivial partition $\{0, \dots, d\} = S \sqcup T$, with $s = |S|$, $t = |T|$, and $s + t = m$. Define $c \in V$ by

$$c_i = \begin{cases} \alpha = \sqrt{\frac{t}{sm}}, & i \in S, \\ \beta = -\sqrt{\frac{s}{tm}}, & i \in T. \end{cases}$$

Then $\sum_i c_i = 0$ and $\|c\| = 1$. Choose $A \in \text{SO}(V)$ with $A^T a = c$. Thus

$$a \cdot A\nu_{ij} = \frac{c_i - c_j}{\sqrt{2}}.$$

Putting $\kappa = (\alpha - \beta)/\sqrt{2}$, the values are 0 within either block and $\pm\kappa$ across the two blocks. Since h is odd, define

$$z_i = \begin{cases} \sqrt{2} h(\kappa), & i \in S, \\ 0, & i \in T. \end{cases}$$

After subtracting its mean, $z \in V$, and

$$h(a \cdot A\nu_{ij}) = \frac{z_i - z_j}{\sqrt{2}} = z \cdot \nu_{ij}.$$

Taking $b = Az$ proves the claim. $\square$

5. THE DIMENSION-FOUR COUNTEREXAMPLE

5.1. A cyclic regular simplex. Identify $\mathbb{R}^4$ with $\mathbb{C}^2$, equipped with the real inner product

$$\langle z, w \rangle_{\mathbb{R}} = \text{Re}(z_1 \bar{w}_1 + z_2 \bar{w}_2).$$

Let $\zeta = e^{2\pi i/5}$ and define

$$q_j = \sqrt{\frac{2}{5}}(\zeta^j, \zeta^{2j}), \quad j = 0, \dots, 4. \quad (12)$$

Lemma 5.1. *The vectors $q_0, \dots, q_4$ are the vertices of a centered regular 4-simplex:*

$$\sum_{j=0}^4 q_j = 0, \quad \langle q_i, q_j \rangle_{\mathbb{R}} = \begin{cases} \frac{4}{5}, & i = j, \\ -\frac{1}{5}, & i \neq j. \end{cases}$$

Consequently

$$\nu_{ij} = \frac{q_i - q_j}{\sqrt{2}} \quad (13)$$

are unit vectors forming an A_4 root system.

Proof. The sum vanishes because both $\sum_j \zeta^j$ and $\sum_j \zeta^{2j}$ vanish. Moreover,

$$\langle q_i, q_j \rangle_{\mathbb{R}} = \frac{2}{5} \text{Re}(\zeta^{i-j} + \zeta^{2(i-j)}).$$

For $i \neq j$ the real part is $-1/2$, so the inner product is $-1/5$; for $i = j$ it is $4/5$. Hence $\|q_i - q_j\|^2 = 2$. The isometry $e_j - \frac{1}{5}\mathbf{1} \mapsto q_j$ identifies this model with the standard one. Composing with an odd permutation of the vertices if necessary makes the isometry orientation-preserving. $\square$

Define

$$W(z) = z_1^2 \bar{z}_2, \quad Q(z) = \text{Re } W(z), \quad R(z) = \text{Im } W(z), \quad H(z) = |z_1|^2 R(z). \quad (14)$$

These are odd real polynomials. Consider the one-parameter subgroup

$$h_\theta(z_1, z_2) = (e^{i\theta} z_1, e^{2i\theta} z_2), \quad J(z_1, z_2) = (iz_1, 2iz_2). \quad (15)$$

Both Q and R are invariant under h_θ .

For $g \in \text{SO}(4)$ put $p_i = gq_i$ and define

$$X(g)_{ij} = \langle p_i, Jp_j \rangle_{\mathbb{R}}, \quad \Lambda_g(Y) = \sum_{i < j} X(g)_{ij} Y_{ij}. \quad (16)$$

The matrix $X(g)$ is skew, and

$$\sum_j X(g)_{ij} = \left\langle p_i, J \sum_j p_j \right\rangle_{\mathbb{R}} = 0,$$

so $X(g) \in \mathcal{C}$. Therefore Λ_g annihilates every exact label. By [theorem 3.4](#),

$$\Lambda_g(Y_Q(g)) = 0, \quad \Lambda_g(Y_R(g)) = 0 \quad \text{for every } g \in \text{SO}(4). \quad (17)$$

The only special calculation needed in the main argument is the following. Its proof is given in [section A](#).

Proposition 5.2 (Algebraic transversality). *If the labels $(Q(g\nu_{ij}))_{i < j}$ are exact, then*

$$\Lambda_g(Y_H(g)) \neq 0. \quad (18)$$

5.2. A compact obstruction lemma.

Lemma 5.3 (Compact obstruction). *Let M be compact, let E be a finite-dimensional normed space, and let $F_0, F_1: M \rightarrow E$ be continuous. Suppose that $x \mapsto \lambda_x \in E^*$ is continuous and*

$$\lambda_x(F_0(x)) = 0 \quad \text{for every } x \in M, \quad (19)$$

$$F_0(x) = 0 \implies \lambda_x(F_1(x)) \neq 0. \quad (20)$$

Then there is $\varepsilon_0 > 0$ such that

$$F_0(x) + \varepsilon F_1(x) \neq 0$$

for every $x \in M$ and every $0 < |\varepsilon| < \varepsilon_0$.

Proof. Let $Z = F_0^{-1}(0)$. By compactness and (20), there is a neighborhood U of Z on which $\lambda_x(F_1(x)) \neq 0$. On the compact set $M \setminus U$,

$$a := \min_{M \setminus U} \|F_0\| > 0.$$

Let $B = \max_M \|F_1\|$. For $0 < |\varepsilon| < a/(B+1)$ the sum cannot vanish on $M \setminus U$. If it vanished at $x \in U$, applying λ_x and using (19) would give $0 = \varepsilon \lambda_x(F_1(x))$, a contradiction. $\square$

5.3. Completion of the proof. Apply [theorem 5.3](#) with

$$M = \text{SO}(4), \quad E = \mathcal{C}, \quad F_0 = \mathcal{T}_Q, \quad F_1 = \mathcal{T}_H, \quad \lambda_g(Y) = \langle X(g), Y \rangle_{\mathcal{E}}.$$

Equation (17) gives $\lambda_g(F_0(g)) = 0$ for all g . If $F_0(g) = 0$, then the Q -labels are exact, and [theorem 5.2](#) gives $\lambda_g(F_1(g)) \neq 0$. Hence, for all sufficiently small nonzero ε ,

$$\mathcal{T}_{Q+\varepsilon H}(g) \neq 0 \quad \text{for every } g \in \text{SO}(4).$$

By [theorem 2.2](#), $Q + \varepsilon H$ is a counterexample.

It remains to obtain the harmonic form stated in [theorem 1.1](#). The cubic R is harmonic, and in $\mathbb{R}^4$ one checks

$$\Delta(|z_1|^2 R) = 12R, \quad \Delta((|z_1|^2 + |z_2|^2)R) = 20R.$$

Thus

$$H_5(z) = \left(|z_1|^2 - \frac{3}{5}(|z_1|^2 + |z_2|^2) \right) R(z) \quad (21)$$

is a homogeneous harmonic quintic. On S^3 , $H_5 = H - \frac{3}{5}R$, and (17) implies

$$\Lambda_g(Y_{H_5}(g)) = \Lambda_g(Y_H(g)).$$

The same compact-obstruction argument therefore applies to $Q + \varepsilon H_5$. Its restriction to S^3 is exactly (2), proving [theorem 1.1](#).

APPENDIX A. PROOF OF ALGEBRAIC TRANSVERSALITY

We prove [theorem 5.2](#). Fix $g \in \text{SO}(4)$ for which the Q -labels are exact, write

$$p_i = gq_i = (a_i, b_i) \in \mathbb{C}^2, \quad a = (a_0, \dots, a_4), \quad b = (b_0, \dots, b_4) \in \mathbb{C}^5,$$

and use coordinatewise multiplication of vectors in $\mathbb{C}^5$. The Hermitian inner product is conjugate-linear in the first variable:

$$\langle x, y \rangle_{\mathbb{C}^5} = \sum_i \bar{x}_i y_i.$$

A.1. Frame identities. The 5×4 real matrix whose rows are the p_i has orthonormal columns, all perpendicular to $\mathbf{1}$. Therefore

$$\sum_i a_i = \sum_i b_i = 0, \tag{22}$$

$$\sum_i a_i^2 = \sum_i b_i^2 = \sum_i a_i b_i = \sum_i a_i \bar{b}_i = 0, \tag{23}$$

$$\sum_i |a_i|^2 = \sum_i |b_i|^2 = 2, \tag{24}$$

$$|a_i|^2 + |b_i|^2 = \frac{4}{5}. \tag{25}$$

Consequently

$$\mathbf{1}, a, b, \bar{a}, \bar{b}$$

are mutually orthogonal in $\mathbb{C}^5$, with squared norms 5, 2, 2, 2, 2. Every $x \in \mathbb{C}^5$ has the expansion

$$x = \frac{\sum_i x_i}{5} \mathbf{1} + \frac{1}{2} \left(\langle a, x \rangle a + \langle b, x \rangle b + \langle \bar{a}, x \rangle \bar{a} + \langle \bar{b}, x \rangle \bar{b} \right). \tag{26}$$

For $u, v \in \mathbb{C}^5$, write

$$(u \wedge v)_{ij} = u_i v_j - v_i u_j.$$

A.2. Cycle equations for the cubic. At the root $g\nu_{ij} = (p_i - p_j)/\sqrt{2}$,

$$W(g\nu_{ij}) = \frac{1}{2\sqrt{2}} (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Ignoring the harmless factor $1/(2\sqrt{2})$, let

$$\tilde{Y}_{ij} = (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Modulo exact matrices, expansion gives

$$\Pi \tilde{Y} \Pi \equiv -a^2 \wedge \bar{b} + 2a \wedge (a\bar{b}). \tag{27}$$

Indeed, the omitted term is $a_i^2 \bar{b}_i - a_j^2 \bar{b}_j$.

Define

$$s = \frac{1}{2} \sum_i |a_i|^2 a_i, \quad \lambda = \frac{1}{2} \sum_i |a_i|^2 \bar{b}_i, \tag{28}$$

$$m = \sum_i a_i^2 \bar{b}_i, \quad n = \sum_i a_i^3, \quad u = \frac{1}{2} \sum_i a_i \bar{b}_i^2, \quad \ell = \frac{1}{2} \sum_i a_i^2 b_i. \tag{29}$$

Using (26),

$$a^2 = sa + \frac{m}{2} b + \frac{n}{2} \bar{a} + \ell \bar{b}, \tag{30}$$

$$a\bar{b} = \lambda a + ub + \frac{m}{2} \bar{a} - s\bar{b}. \tag{31}$$

Substitution into (27) yields

$$\omega := \Pi \tilde{Y} \Pi = 2u a \wedge b + m a \wedge \bar{a} - 3s a \wedge \bar{b} - \frac{m}{2} b \wedge \bar{b} - \frac{n}{2} \bar{a} \wedge \bar{b}. \quad (32)$$

Exactness of the real Q -labels is precisely $\operatorname{Re} \omega = 0$. The six wedges of the complex basis $a, b, \bar{a}, \bar{b}$ are linearly independent. Comparing (32) with its conjugate gives

$$s = 0, \quad m \in \mathbb{R}, \quad 4u = \bar{n}. \quad (33)$$

Replacing g by $h_\theta g$ changes neither the values of Q and H nor the matrix $X(g)$, while $\lambda \mapsto e^{-2i\theta} \lambda$. We may therefore assume

$$\lambda = t \geq 0. \quad (34)$$

In addition define

$$v = \sum_i a_i b_i^2, \quad k = \sum_i b_i^3. \quad (35)$$

A.3. The coordinatewise multiplication table. Expanding each product in the orthogonal basis $\mathbf{1}, a, b, \bar{a}, \bar{b}$ by (26), and using (25) and (33), gives

$$\begin{aligned} a\bar{a} &= \frac{2}{5}\mathbf{1} + tb + t\bar{b}, & b\bar{b} &= \frac{2}{5}\mathbf{1} - tb - t\bar{b}, \\ a^2 &= \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b}, & \bar{a}^2 &= \frac{m}{2}\bar{b} + \frac{\bar{n}}{2}a + \bar{\ell}b, \\ a\bar{b} &= ta + \frac{\bar{n}}{4}b + \frac{m}{2}\bar{a}, & \bar{a}b &= t\bar{a} + \frac{n}{4}\bar{b} + \frac{m}{2}a, \\ ab &= ta + \ell\bar{a} + \frac{v}{2}\bar{b}, & \bar{a}\bar{b} &= t\bar{a} + \bar{\ell}a + \frac{\bar{v}}{2}b, \\ b^2 &= \frac{n}{4}a - tb + \frac{v}{2}\bar{a} + \frac{k}{2}\bar{b}, & \bar{b}^2 &= \frac{\bar{n}}{4}\bar{a} - t\bar{b} + \frac{\bar{v}}{2}a + \frac{\bar{k}}{2}b. \end{aligned} \quad (36)$$

For example, the coefficient of b in $a\bar{a}$ is $\frac{1}{2} \sum_i \bar{b}_i |a_i|^2 = t$, and the other coefficients are obtained in the same way.

Coordinatewise multiplication is associative. Comparing coefficients in (36) gives the following necessary equations:

$$3mn = 4tv, \quad (E1)$$

$$\bar{n}v + 16\ell t + 8mt = 0, \quad (E2)$$

$$-2km + 24\ell t - n^2 = 0, \quad (E3)$$

$$-kt + \ell m + 4t^2 = 0, \quad (E4)$$

$$\bar{n}k + mn + 8tv = 0, \quad (E5)$$

$$20|\ell|^2 + 5m^2 + 5|n|^2 - 40t^2 = 8, \quad (E6)$$

$$16|\ell|^2 - 4m^2 - |n|^2 + 4|v|^2 = 0, \quad (E7)$$

$$20|\ell|^2 + 60t^2 + 5|v|^2 = 8. \quad (E8)$$

More precisely, (E1)–(E3) are the $a, b, \bar{b}$ coefficients of $(a^2)b = a(ab)$; (E4)–(E5) are the a, b coefficients of $(ab)b = a(b^2)$; and (E6), (E7), (E8) are respectively the indicated coefficients of

$$(a^2)\bar{a} = a(a\bar{a}), \quad (ab)\bar{a} = a(b\bar{a}), \quad (ab)\bar{b} = a(b\bar{b}).$$

A.4. Classification of the possible cubic zeros. We extract the consequences of (E1)–(E8) needed below.

Case I: $t = 0$. Equation (E1) gives $mn = 0$. If $m = 0$, then (E3) gives $n = 0$, while (E6)–(E7) become

$$20|\ell|^2 = 8, \quad 16|\ell|^2 + 4|v|^2 = 0,$$

a contradiction. Hence $m \neq 0$ and $n = 0$. Equations (E4) and (E3) give $\ell = k = 0$, and then

$$m^2 = \frac{8}{5}, \quad |v|^2 = m^2. \quad (37)$$

Case II: $t > 0$ and $n = 0$. Equation (E1) gives $v = 0$, and (E2) gives

$$\ell = -\frac{m}{2}. \quad (38)$$

Here $m \neq 0$, since otherwise (36) would give $a^2 = 0$. Equations (E3) and (E4) give $k = -6t$ and $m^2 = 20t^2$. Finally (E8) yields

$$t^2 = \frac{1}{20}, \quad m^2 = 1. \quad (39)$$

Case III: $t > 0$ and $n \neq 0$. First $m \neq 0$, since otherwise (E1), (E2), and (E3) successively give $v = 0$, $\ell = 0$, and $n = 0$. Put

$$q = \frac{m}{t}, \quad h = \frac{|n|^2}{t^2} > 0. \quad (40)$$

Equations (E1) and (E2) give

$$v = \frac{3mn}{4t}, \quad \ell = -\frac{m(32 + 3h)}{64}. \quad (41)$$

In particular, ℓ is real. Equation (E5) gives

$$k = -7m\frac{n}{n}.$$

Equation (E4) shows that k is real, so

$$\sigma := \frac{n}{\bar{n}} \in \{1, -1\}, \quad k = -7m\sigma, \quad n^2 = \sigma |n|^2. \quad (42)$$

After division by appropriate powers of t , equations (E4) and (E3) become

$$3hq^2 + 32q^2 - 448\sigma q - 256 = 0, \quad (43)$$

$$-9hq - 8\sigma h + 112\sigma q^2 - 96q = 0. \quad (44)$$

Eliminating h gives

$$(q - 4)(q + 2)(21q^2 + 42q + 16) = 0 \quad (\sigma = 1), \quad (45)$$

$$(q + 4)(q - 2)(21q^2 - 42q + 16) = 0 \quad (\sigma = -1). \quad (46)$$

The roots $q = -2\sigma$ give $h = -64$ and are impossible. One root of each quadratic gives $h = (-74 - 14\sqrt{105})/3 < 0$. The candidates $q = 4\sigma$ give $h = 32$; then (41) and (E1) imply

$$\frac{|\ell|^2}{t^2} = 64, \quad \frac{|v|^2}{t^2} = 288,$$

and the left side of (E7), divided by t^2 , is 2080, a contradiction. The only remaining possibility is

$$q = -\sigma \left(1 - \frac{\sqrt{105}}{21}\right), \quad h = \frac{-74 + 14\sqrt{105}}{3}. \quad (47)$$

Equation (E6) then gives

$$t^2 = \frac{7(5\sqrt{105} - 33)}{1440}, \quad m^2 = \frac{49\sqrt{105} - 477}{1080}. \quad (48)$$

These three cases exhaust every actual cubic zero, because every such zero satisfies the necessary equations (E1)–(E8).

A.5. Evaluation of the quintic functional. For the actual root $g\nu_{ij} = (p_i - p_j)/\sqrt{2}$,

$$|z_1|^2 W(z) = \frac{1}{4\sqrt{2}}(a_i - a_j)^3(\bar{a}_i - \bar{a}_j)(\bar{b}_i - \bar{b}_j). \quad (49)$$

Define

$$\Sigma = \sum_{i < j} X(g)_{ij}(a_i - a_j)^3(\bar{a}_i - \bar{a}_j)(\bar{b}_i - \bar{b}_j). \quad (50)$$

Then

$$\Lambda_g(Y_H(g)) = \frac{1}{4\sqrt{2}} \text{Im } \Sigma. \quad (51)$$

In wedge notation,

$$X(g) = \frac{i}{2}(\bar{a} \wedge a + 2\bar{b} \wedge b). \quad (52)$$

Modulo an exact matrix, the degree-five edge expression in (50) is

$$\begin{aligned} F \equiv & \bar{b} \wedge (a^3 \bar{a}) + \bar{a} \wedge (a^3 \bar{b}) - (\bar{a} \bar{b}) \wedge a^3 \\ & + 3a \wedge (a^2 \bar{a} \bar{b}) - 3(a \bar{b}) \wedge (a^2 \bar{a}) \\ & - 3(a \bar{a}) \wedge (a^2 \bar{b}) + 3(a \bar{a} \bar{b}) \wedge a^2. \end{aligned} \quad (53)$$

The omitted exact term is $(a^3 \bar{a} \bar{b}) \wedge \mathbf{1}$, which is annihilated by $X(g)$. For the bilinear dot product $x \cdot y = \sum_i x_i y_i$,

$$\sum_{i < j} (u \wedge v)_{ij}(x \wedge y)_{ij} = (u \cdot x)(v \cdot y) - (u \cdot y)(v \cdot x). \quad (54)$$

Every dot product in (54) is five times the coefficient of $\mathbf{1}$ in the corresponding coordinatewise product. Substituting (36), (52), and (53) gives

$$\Sigma = -\frac{i}{80} \mathcal{B}_{\text{raw}}, \quad (55)$$

where

$$\begin{aligned} \mathcal{B}_{\text{raw}} = & -960\bar{k} \ell t + 160|\ell|^2 m + 320\ell t^2 - 120m^3 \\ & - 75m|n|^2 - 800mt^2 + 100m|v|^2 + 192m - 160nt\bar{v}. \end{aligned} \quad (56)$$

In all three cases above, ℓ is real. The conjugates of (E4) and (E1) give

$$\bar{k}t = \ell m + 4t^2, \quad nt\bar{v} = \frac{3}{4}m|n|^2. \quad (57)$$

Using (57), followed by (E6) and (E8), reduces (56) to

$$\mathcal{B} = m(180m^2 + 105|n|^2 - 4400t^2 - 128) - 3520\ell t^2. \quad (58)$$

The three cases now give

case	$\mathcal{B}$
I	$160m$
II	$-80m$
III	$60m(75 - 7\sqrt{105})$

(59)

Each value is nonzero: in every case $m \neq 0$, and $75 - 7\sqrt{105} > 0$. Since $\mathcal{B}$ is real, (51) and (55) yield

$$\Lambda_g(Y_H(g)) = -\frac{\mathcal{B}}{320\sqrt{2}} \neq 0.$$

This proves [theorem 5.2](#).

Remark A.1 (Symbolic verification). The wedge expansion (53), the extraction of (E1)–(E8), the resultants (45)–(46), and the simplification (58) are polynomial identities. An accompanying exact-arithmetic SymPy script checks each of them independently.

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